

higher, at 0.42% per month (5% per year) but is much more statistically significant with a t -statistic of 4.37. Note that both *BAB* and *VOL* have significant value tilts (positive *HML* loadings) and momentum tilts (positive *UMD* loadings). The big difference is that the *BAB* factor carries a negative *SMB* loading, whereas it is positive for the *VOL* factor. That is, the beta anomaly manifests more in small stocks. In contrast, the volatility anomaly is more pervasive in large stocks, which are usually easier to trade because they are more liquid.

So should you do low volatility, or should you do low beta? This is not an either-or choice. You should do both.

4.5 EXPLANATIONS

We are still searching for a comprehensive explanation for the risk anomaly. In my opinion, the true explanation is a combination of all of the explanations listed below, plus potentially others being developed.

Data Mining

Some papers in the literature rightfully point out some data mining concerns with the original results in Ang et al. (2006). There is some sensitivity in the results to different portfolio weighting schemes and illiquidity effects.²⁹ For the most part, however, the low-risk anomaly is fairly robust. A recent survey article by Chen et al. (2012) argues that “idiosyncratic volatility is a common stock phenomenon” and is not due to microstructure or liquidity biases.

The best argument against data mining is that the low-risk effect is seen in many other contexts. Ang et al. (2006) show that the effect appears during recessions and expansions and during stable and volatile periods. Ang et al. (2009) show that it takes place in international stock markets. Frazzini and Pedersen (2011) show that low-beta portfolios have high Sharpe ratios in U.S. stocks, international stocks, Treasury bonds, corporate bonds cut by different maturities and credit ratings, credit derivative markets, commodities, and foreign exchange. Cao and Han (2013) and Blitz and de Groot (2013) show that the low-risk phenomenon even shows up in option and commodity markets, respectively. Low risk is pervasive.

Leverage Constraints

Many investors are leverage constrained—they wish to take on more risk but are unable to take on more leverage.³⁰ Since they cannot borrow, they do the next best thing—they hold stocks with “built-in” leverage, like high-beta stocks. Investors bid up the price of high-beta stocks until the shares are overpriced and deliver low

²⁹ See Bali and Cakici (2008) and Han and Lesmond (2011), respectively.

³⁰ Black (1972) was the first to develop a theory of the CAPM for when investors cannot lever. Frazzini and Pedersen (2011) apply a leverage-constraint story to explain the low-beta anomaly.